

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397800
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Raikar; Akshay Pai et al.

Systems and methods for automatically dispensing road markings for autonomous vehicle signaling

Abstract

An autonomous vehicle comprises a sensor, a plurality of paint dispensers, and one or more processors. The processors can be configured to collect, using the sensor, data regarding an environment surrounding the autonomous vehicle as the autonomous vehicle is moving on a surface; detect an event in the environment surrounding the autonomous vehicle from the collected data; determine an event type of the event based on the collected data; select a paint dispenser from the plurality of paint dispensers based on the determined event type; and automatically dispense paint from the selected paint dispenser onto the surface.

Inventors: Raikar; Akshay Pai (Blacksburg, VA), Fox-Rabinovitz; Joseph (Austin, TX)

Applicant: TORC Robotics, Inc. (Blacksburg, VA)

Family ID: 1000008779334

Assignee: TORC ROBOTICS, INC. (Blacksburg, VA)

Appl. No.: 18/219609

Filed: July 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250010864 A1	Jan. 09, 2025

Publication Classification

Int. Cl.: B60W40/06 (20120101); B60W60/00 (20200101); E01C23/16 (20060101)

U.S. Cl.:

CPC **B60W40/06** (20130101); **B60W60/00** (20200201); **E01C23/16** (20130101);
B60W2552/00 (20200201)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10976746	12/2020	Hendricks et al.	N/A	N/A
11828610	12/2022	Yeung	N/A	N/A
2018/0079359	12/2017	Park	N/A	G08G 1/163
2022/0284223	12/2021	Lev	N/A	G08G 1/09623
2023/0384096	12/2022	Salter et al.	N/A	N/A
2023/0419606	12/2022	Stenson et al.	N/A	N/A

Primary Examiner: Patel; Hitesh

Assistant Examiner: Yang; Wenyuan

Attorney, Agent or Firm: Armstrong Teasdale LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to vehicles and, more specifically, to systems and methods for dispensing road markings outside of a visible spectrum by an autonomous vehicle for autonomous vehicle signaling.

BACKGROUND

(2) The use of autonomous vehicles has become increasingly prevalent in recent years, with the potential for numerous benefits, such as improved safety, reduced traffic congestion, and increased mobility for people with disabilities. Roadways can include highly variable conditions, with traffic flow, congestion, pull overs, crashes, construction zones, and bus stops contributing to non-permanent, irregular conditions. Autonomous vehicles can collect a large variety of data regarding the changing environment surrounding the autonomous vehicles. However, given the limited fields of view of the sensors on the autonomous vehicles, autonomous vehicles may detect irregular conditions too late to react to the irregular conditions or may fail to detect the irregular conditions altogether.

SUMMARY

(3) Roadway systems across the world experience events that cause changes to the conditions of a road. These events may include man generated events (e.g., construction, crashes, pull over, rush hour, etc.), nature-based events (e.g., hurricanes, tornadoes, floods, etc.), or other events that may produce non-permanent, irregular conditions on the road. As a vehicle passes through a road segment affected by one or more of these events, the vehicle may determine the condition (or the event that caused the condition) of the road segment. Subsequent vehicles may benefit from

obtaining information about the road segment before reaching the road segment. For example, a first vehicle may approach a construction zone and may determine to merge into a neighboring lane to avoid the construction zone. A second vehicle behind the first vehicle may improve route efficiency and avoid congestion by merging into the neighboring lane before approaching the construction zone. In this way, route planning for autonomous vehicles may be improved.

(4) With the deployment of autonomous vehicles on public roads, and due to the technical nature of the autonomous vehicles, improvements to information sharing between vehicles about non-permanent conditions are possible. For example, an automated (e.g., autonomous) vehicle system implementing the systems and methods described herein may result in improved communication between vehicles. For example, a computer of the autonomous vehicle system may operate to collect, using a sensor of the autonomous vehicle, data regarding an environment surrounding the autonomous vehicle as the autonomous vehicle is moving on a surface. The autonomous vehicle may detect an event (e.g., a car crash, excessive traffic, a wet road, etc.) in the environment surrounding the autonomous vehicle from the collected data. The autonomous vehicle may determine an event type of the detected event based on the collected data. The autonomous vehicle may select a paint dispenser from multiple paint dispensers included in the autonomous vehicle based on the determined event type. The autonomous vehicle may automatically dispense paint from the selected paint dispenser onto the surface to convey information about the event to other vehicles.

(5) To determine the event type, the computer may utilize one or more algorithms. For example, the computer may analyze image or video data depicting the event. The computer may use a neural network model or other image processing algorithm to determine the type of the event. For example, event types may include temporal designations (e.g., a short duration event, a long duration event, etc.), characteristic designations (e.g., construction zone, road debris, etc.), emergency designations (e.g., high emergency, low emergency), or any other categorization of event type.

(6) The computer may select the paint dispenser based on the event type. For example, the computer may detect an event while driving on the road. Responsive to doing so, the computer may determine a first event type associated with a first duration of time. In an example, the first event type may be a car crash that is associated with a short duration of time (e.g., 1 hour, 2 hours, etc.). The computer may select a first paint dispenser associated with the event type and the short duration of time. The paint dispenser may dispense paint that indicates the event type and the short duration of time. For example, the computer may be configured with a database including paint types (e.g., color, wavelength, size, shape, etc.) corresponding to different event types and durations. The paint may be non-permanent such that the paint disappears (e.g., dissolves, is undetectable by the sensor) after the duration of time transpires. In some cases, the database may include paint types corresponding to navigational information that indicate navigational actions. A computer of a second autonomous vehicle may detect, via a sensor, the paint and determine the paint type. The computer may operate the second autonomous vehicle to perform a navigational action responsive to detection of the paint and identifying the paint type.

(7) Advantageously, by performing the methods or adopting the systems as described herein, vehicles can provide road specific information associated with non-permanent conditions to other vehicles. This may result in increased efficiency for route planning, reduced congestion and latency in completing a route, and increased communication between vehicles (e.g., in areas without communication via a network, without wireless communication between the vehicles), among other advantages.

(8) While the examples described herein are described in the context of autonomous vehicles, any vehicle with a computer that can detect the tracking tags may utilize the systems and methods as described herein.

(9) In at least one aspect, the present disclosure describes an autonomous vehicle. The autonomous

vehicle can include one or more sensors, multiple paint dispensers and one or more processors coupled with the one or more sensors and the multiple paint dispensers. The one or more processors can be configured to collect, using the sensors, data regarding an environment surrounding the autonomous vehicle as the autonomous vehicle is moving on a surface; detect an event in the environment surrounding the autonomous vehicle from the collected data; determine an event type of the event based on the collected data; select a paint dispenser from the plurality of paint dispensers based on the determined event type; and automatically dispense paint from the selected paint dispenser onto the surface.

(10) In another aspect, the present disclosure describes a method for dispensing paint onto a surface. The method can include collecting, using a sensor, data regarding an environment surrounding a vehicle as the vehicle is moving on the surface; detecting an event in the environment surrounding the vehicle from the collected data; determining an event type of the event based on the collected data; selecting a paint dispenser from a plurality of paint dispensers based on the determined event type, each dispenser of the plurality of paint dispensers corresponding to a different type of event and comprising paint of a different spectral band, wherein paint from each of the plurality of paint dispensers is outside of a visible spectrum; and automatically dispensing paint from the selected paint dispenser onto the surface.

(11) In another aspect, the present disclosure describes a controller. The controller can include one or more processors configured to collect, using the sensor, data regarding an environment surrounding an autonomous vehicle as the autonomous vehicle is moving on a surface; detect an event in the environment surrounding the autonomous vehicle from the collected data; determine an event type of the event based on the collected data; select a paint dispenser from a plurality of paint dispensers based on the determined event type, each dispenser of the plurality of paint dispensers corresponding to a different type of event and comprising paint of a different spectral band, wherein paint from each of the plurality of paint dispensers is outside of a visible spectrum; and automatically dispense paint from the selected paint dispenser onto the surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

(2) FIG. 1 is a bird's-eye view of a roadway including a schematic representation of a vehicle and aspects of an autonomy system of the vehicle, according to an embodiment.

(3) FIG. 2 is a schematic of an autonomy system of a vehicle, according to an embodiment.

(4) FIGS. 3-4 are bird's-eye views of roadways that support dispensing road markings outside of a visible spectrum by an autonomous vehicle, according to embodiments.

(5) FIG. 5 is a method for dispensing road markings outside of a visible spectrum by an autonomous vehicle, according to an embodiment.

DETAILED DESCRIPTION

(6) The following detailed description describes various features and functions of the disclosed systems and methods with reference to the accompanying figures. In the figures, similar components are identified using similar symbols, unless otherwise contextually dictated. The exemplary system(s) and method(s) described herein are not limiting and it may be readily understood that certain aspects of the disclosed systems and methods can be variously arranged and combined, all of which arrangements and combinations are contemplated by this disclosure.

(7) Referring to FIG. 1, the present disclosure relates to autonomous vehicles, such as an autonomous vehicle **102** having an autonomy system **114**. The autonomy system **114** of the vehicle

102 may be completely autonomous (fully autonomous) or semi-autonomous. In one example, the autonomous system **114** can operate under Level 5 autonomy (e.g., full driving automation), Level 4 autonomy (e.g., high driving automation), or Level 3 autonomy (e.g., conditional driving automation). As used herein the term “autonomous” includes both fully autonomous and semi-autonomous. The present disclosure sometimes refers to autonomous vehicles as ego vehicles. The autonomy system **114** may be structured on at least three aspects of technology: (1) perception, (2) maps/localization, and (3) behaviors planning and control. The function of the perception aspect is to sense an environment surrounding the vehicle **102** and interpret the environment. To interpret the surrounding environment, a perception module **116** or engine in the autonomy system **114** of the vehicle **102** may identify and classify objects or groups of objects in the environment. For example, a perception module **116** may be associated with various sensors (e.g., light detection and ranging (LiDAR), camera, radar, etc.) of the autonomy system **114** and may identify one or more objects (e.g., pedestrians, vehicles, debris, etc.) and features of the roadway (e.g., lane lines) around the vehicle **102**, and classify the objects in the road distinctly.

(8) The maps/localization aspect of the autonomy system **114** may be configured to determine where on a pre-established digital map the vehicle **102** is currently located. One way to do this is to sense the environment surrounding the vehicle **102** (e.g., via the perception module **116**), such as by detecting vehicles (e.g., a vehicle **104**) or other objects (e.g., traffic lights, speed limit signs, pedestrians, signs, road markers, energy supply stations, etc.) from data collected via the sensors of the autonomy system **114**, and to correlate features of the sensed environment with details (e.g., digital representations of the features of the sensed environment) on the digital map.

(9) Once the systems on the vehicle **102** have determined the location of the vehicle **102** with respect to the digital map features (e.g., location on the roadway, upcoming intersections, road signs, etc.), the vehicle **102** can plan and execute maneuvers and/or routes with respect to the features of the digital map. The behaviors, planning, and control aspects of the autonomy system **114** may be configured to make decisions about how the vehicle **102** should move through the environment to get to the goal or destination of the vehicle **102**. The autonomy system **114** may consume information from the perception and maps/localization modules to know where the vehicle **102** is relative to the surrounding environment and what other objects and traffic actors are doing.

(10) FIG. 1 further illustrates an environment **100** for modifying one or more actions of the vehicle **102** using the autonomy system **114**. The vehicle **102** is capable of communicatively coupling to a remote server **122** via a network **120**. The vehicle **102** may not necessarily connect with the network **120** or the server **122** while it is in operation (e.g., driving down the roadway). That is, the server **122** may be remote from the vehicle, and the vehicle **102** may deploy with all the necessary perception, localization, and vehicle control software and data necessary to complete a mission of the vehicle **102** fully autonomously or semi-autonomously.

(11) While this disclosure refers to a vehicle **102** as the autonomous vehicle, it is understood that the vehicle **102** could be any type of vehicle including a truck (e.g., a tractor trailer), an automobile, a mobile industrial machine, etc. While the disclosure will discuss a self-driving or driverless autonomous system, it is understood that the autonomous system could alternatively be semi-autonomous having varying degrees of autonomy or autonomous functionality. While the perception module **116** is depicted as being located at the front of the vehicle **102**, the perception module **116** may be a part of a perception system with various sensors placed at different locations throughout the vehicle **102** (e.g., a front side of the vehicle **102**, an energy input side of the vehicle).

(12) FIG. 2 illustrates an example schematic of an autonomy system **250** of a vehicle **200**, according to some embodiments. The autonomy system **250** may be the same as or similar to the autonomy system **114**. The vehicle **200** may be the same as or similar to the vehicle **102**. The autonomy system **250** may include a perception system including a camera system **220**, a LiDAR

system **222**, a radar system **232**, a Global Navigation Satellite System (GNSS) receiver **208**, an inertial measurement unit (IMU) **224**, and/or a perception module **202**. The autonomy system **250** may further include a transceiver **226**, a processor **210**, a memory **214**, a mapping/localization module **204**, and a vehicle control module **206**. The various systems may serve as inputs to and receive outputs from various other components of the autonomy system **250**. In other examples, the autonomy system **250** may include more, fewer, or different components or systems, and each of the components or system(s) may include more, fewer, or different components. Additionally, the systems and components shown may be combined or divided in various ways. As shown in FIG. 1, the perception systems aboard the autonomous vehicle may help the vehicle **102** perceive the vehicle **102**'s environment out to a perception area **118**. The actions of the vehicle **102** may depend on the extent of the perception area **118**. It is to be understood that the perception area **118** is an example area, and the practical area may be greater than or less than what is depicted.

(13) The camera system **220** of the perception system may include one or more cameras mounted at any location on the vehicle **102**, which may be configured to capture images of the environment surrounding the vehicle **102** in any aspect or field of view (FOV). The FOV can have any angle or aspect such that images of the areas ahead of, to the side, and behind the vehicle **102** may be captured. In some embodiments, the FOV may be limited to particular areas around the vehicle **102** (e.g., forward of the vehicle **102**, at the side of the vehicle **102**) or may surround 360 degrees of the vehicle **102**. In some embodiments, the image data generated by the camera system(s) **220** may be sent to the perception module **202** and stored, for example, in memory **214**.

(14) The LiDAR system **222** may include a laser generator and a detector and can send and receive LiDAR signals. A LiDAR signal can be emitted to and received from any direction such that LiDAR point clouds (or "LiDAR images") of the areas ahead of, to the side, and behind the vehicle **200** can be captured and stored as LiDAR point clouds. In some embodiments, the vehicle **200** may include multiple LiDAR systems and point cloud data from the multiple systems may be stitched together.

(15) The radar system **232** may estimate strength or effective mass of an object, as objects made out of paper or plastic may be weakly detected. The radar system **232** may be based on 24 GHz, 77 GHz, or other frequency radio waves. The radar system **232** may include short-range radar (SRR), mid-range radar (MRR), or long-range radar (LRR). One or more sensors may emit radio waves, and a processor may process received reflected data (e.g., raw radar sensor data) from the emitted radio waves.

(16) In some embodiments, the system inputs from the camera system **220**, the LiDAR system **222**, and the radar system **232** may be fused (e.g., in the perception module **202**). The LiDAR system **222** may include one or more actuators to modify a position and/or orientation of the LiDAR system **222** or components thereof. The LiDAR system **222** may be configured to use ultraviolet (UV), visible, or infrared light to image objects and can be used with a wide range of targets. In some embodiments, the LiDAR system **222** can be used to map physical features of an object with high resolution (e.g., using a narrow laser beam). In some examples, the LiDAR system **222** may generate a point cloud and the point cloud may be rendered to visualize the environment surrounding the vehicle **200** (or object(s) therein). In some embodiments, the point cloud may be rendered as one or more polygon(s) or mesh model(s) through, for example, surface reconstruction. Collectively, the radar system **232**, the LiDAR system **222**, and the camera system **220** may be referred to herein as "imaging systems."

(17) The GNSS receiver **208** may be positioned on the vehicle **200** and may be configured to determine a location of the vehicle **200** via GNSS data, as described herein. The GNSS receiver **208** may be configured to receive one or more signals from a global navigation satellite system (GNSS) (e.g., a GPS) to localize the vehicle **200** via geolocation. The GNSS receiver **208** may provide an input to and otherwise communicate with the mapping/localization module **204** to, for example, provide location data for use with one or more digital maps, such as an HD map (e.g., in a

vector layer, in a raster layer or other semantic map, etc.). In some embodiments, the GNSS receiver **208** may be configured to receive updates from an external network.

(18) The IMU **224** may be an electronic device that measures and reports one or more features regarding the motion of the vehicle **200**. For example, the IMU **224** may measure a velocity, acceleration, angular rate, and/or an orientation of the vehicle **200** or one or more of the vehicle **200**'s individual components using a combination of accelerometers, gyroscopes, and/or magnetometers. The IMU **224** may detect linear acceleration using one or more accelerometers and rotational rate using one or more gyroscopes. In some embodiments, the IMU **224** may be communicatively coupled to the GNSS receiver **208** and/or the mapping/localization module **204** to help determine a real-time location of the vehicle **200** and predict a location of the vehicle **200** even when the GNSS receiver **208** cannot receive satellite signals.

(19) The transceiver **226** may be configured to communicate with one or more external networks **260** via, for example, a wired or wireless connection in order to send and receive information (e.g., to a remote server **270**). The wireless connection may be a wireless communication signal (e.g., Wi-Fi, cellular, LTE, 5G, etc.). In some embodiments, the transceiver **226** may be configured to communicate with external network(s) via a wired connection, such as, for example, during initial installation, testing, or service of the autonomy system **250** of the vehicle **200**. A wired/wireless connection may be used to download and install various lines of code in the form of digital files (e.g., HD digital maps), executable programs (e.g., navigation programs), and other computer-readable code that may be used by the system **250** to navigate the vehicle **200** or otherwise operate the vehicle **200**, either fully autonomously or semi-autonomously.

(20) The processor **210** of autonomy system **250** may be embodied as one or more of a data processor, a microcontroller, a microprocessor, a digital signal processor, a logic circuit, a programmable logic array, or one or more other devices for controlling the autonomy system **250** in response to one or more of the system inputs. The autonomy system **250** may include a single microprocessor or multiple microprocessors that may include means for controlling the vehicle **200** to move (e.g., switch lanes) and monitoring and detecting other vehicles. Numerous commercially available microprocessors can be configured to perform the functions of the autonomy system **250**. It should be appreciated that the autonomy system **250** could include a general machine controller capable of controlling numerous other machine functions. Alternatively, a special-purpose machine controller could be provided. Further, the autonomy system **250**, or portions thereof, may be located remote from the system **250**. For example, one or more features of the mapping/localization module **204** could be located remote to the vehicle **200**. Various other known circuits may be associated with the autonomy system **250**, including signal-conditioning circuitry, communication circuitry, actuation circuitry, and other appropriate circuitry.

(21) The memory **214** of the autonomy system **250** may store data and/or software routines that may assist the autonomy system **250** in performing autonomy system **250**'s functions, such as the functions of the perception module **202**, the mapping/localization module **204**, the vehicle control module **206**, a road signal module **230**, and the method **500** described herein with respect to FIG. 5. Further, the memory **214** may also store data received from various inputs associated with the autonomy system **250**, such as perception data from the perception system.

(22) As noted above, the perception module **202** may receive input from the various sensors, such as the camera system **220**, the LiDAR system **222**, the GNSS receiver **208**, and/or the IMU **224** (collectively "perception data") to sense an environment surrounding the vehicle **200** and interpret it. To interpret the surrounding environment, the perception module **202** (or "perception engine") may identify and classify objects or groups of objects in the environment. For example, the vehicle **102** may use the perception module **202** to identify one or more objects (e.g., pedestrians, vehicles, debris, etc.) or features of the roadway **106** (e.g., intersections, road signs, lane lines, etc.) before or beside a vehicle and classify the objects in the road. In some embodiments, the perception module **202** may include an image classification function and/or a computer vision function.

(23) The system **250** may collect perception data. The perception data may represent the perceived environment surrounding the vehicle, for example, and may be collected using aspects of the perception system described herein. The perception data can come from, for example, one or more of the LiDAR system, the camera system, the radar system and various other externally-facing sensors and systems on board the vehicle (e.g., the GNSS receiver, etc.). For example, in vehicles having a sonar or radar system, the sonar and/or radar systems may collect perception data. As the vehicle **102** travels along the roadway **106**, the system **250** may continually receive data from the various systems on the vehicle **102**. In some embodiments, the system **250** may receive data periodically and/or continuously. With respect to FIG. **1**, the vehicle **102** may collect perception data that indicates the presence of the lane line **110** (e.g., in order to determine the lanes **108** and **112**). Additionally, the detection systems may detect the vehicle **104** and monitor the vehicle **104** to estimate various properties of the vehicle **104** (e.g., proximity, speed, behavior, flashing light, etc.). The properties of the vehicle **104** may be stored as timeseries data in which timestamps indicate the times in which the different properties were measured or determined. The features may be stored as points (e.g., vehicles, signs, small landmarks, etc.), lines (e.g., lane lines, road edges, etc.), or polygons (e.g., lakes, large landmarks, etc.) and may have various properties (e.g., style, visible range, refresh rate, etc.), which properties may control how the system **250** interacts with the various features.

(24) The image classification function may determine the features of an image (e.g., a visual image from the camera system **220** and/or a point cloud from the LiDAR system **222**). The image classification function can be any combination of software agents and/or hardware modules able to identify image features and determine attributes of image parameters in order to classify portions, features, or attributes of an image. The image classification function may be embodied by a software module that may be communicatively coupled to a repository of images or image data (e.g., visual data and/or point cloud data) which may be used to determine objects and/or features in real-time image data captured by, for example, the camera system **220** and the LiDAR system **222**. In some embodiments, the image classification function may be configured to classify features based on information received from only a portion of the multiple available sources. For example, in the case that the captured visual camera data includes images that may be blurred, the system **250** may identify objects based on data from one or more of the other systems (e.g., the LiDAR system **222**) that does not include the image data.

(25) The computer vision function may be configured to process and analyze images captured by the camera system **220** and/or the LiDAR system **222** or stored on one or more modules of the autonomy system **250** (e.g., in the memory **214**), to identify objects and/or features in the environment surrounding the vehicle **200** (e.g., lane lines). The computer vision function may use, for example, an object recognition algorithm, video tracing, one or more photogrammetric range imaging techniques (e.g., a structure from motion (SfM) algorithms), or other computer vision techniques. The computer vision function may be configured to, for example, perform environmental mapping and/or track object vectors (e.g., speed and direction). In some embodiments, objects or features may be classified into various object classes using the image classification function, for instance, and the computer vision function may track the one or more classified objects to determine aspects of the classified object (e.g., aspects of the vehicle **200**'s motion, size, etc.)

(26) The mapping/localization module **204** receives perception data that can be compared to one or more digital maps stored in the mapping/localization module **204** to determine where the vehicle **200** is in the world and/or where the vehicle **200** is on the digital map(s). In particular, the mapping/localization module **204** may receive perception data from the perception module **202** and/or from the various sensors sensing the environment surrounding the vehicle **200** and correlate features of the sensed environment with details (e.g., digital representations of the features of the sensed environment) on the one or more digital maps. The digital map may have various levels of

detail and can be, for example, a raster map, a vector map, etc. The digital maps may be stored locally on the vehicle **200** and/or stored and accessed remotely.

(27) The vehicle control module **206** may control the behavior and maneuvers of the vehicle **200**. For example, once the systems on the vehicle **200** have determined the vehicle **200**'s location with respect to map features (e.g., intersections, road signs, lane lines, etc.) the vehicle **200** may use the vehicle control module **206** and the vehicle **200**'s associated systems to plan and execute maneuvers and/or routes with respect to the features of the environment. The vehicle control module **206** may make decisions about how the vehicle **200** will move through the environment to get to the vehicle **200**'s goal or destination as it completes the vehicle **200**'s mission. The vehicle control module **206** may consume information from the perception module **202** and the mapping/localization module **204** to know where it is relative to the surrounding environment and what other traffic actors are doing.

(28) The vehicle control module **206** may be communicatively and operatively coupled to a plurality of vehicle operating systems and may execute one or more control signals and/or schemes to control operation of the one or more operating systems, for example, the vehicle control module **206** may control one or more of a vehicle steering system, a propulsion system, and/or a braking system. The propulsion system may be configured to provide powered motion for the vehicle **200** and may include, for example, an engine/motor, an energy source, a transmission, and wheels/tires and may be coupled to and receive a signal from a throttle system, for example, which may be any combination of mechanisms configured to control the operating speed and acceleration of the engine/motor and thus, the speed/acceleration of the vehicle **200**. The steering system may be any combination of mechanisms configured to adjust the heading or direction of the vehicle **200**. The brake system may be, for example, any combination of mechanisms configured to decelerate the vehicle **200** (e.g., friction braking system, regenerative braking system, etc.) The vehicle control module **206** may be configured to avoid obstacles in the environment surrounding the vehicle **200** and may be configured to use one or more system inputs to identify, evaluate, and modify a vehicle trajectory. The vehicle control module **206** is depicted as a single module, but can be any combination of software agents and/or hardware modules able to generate vehicle control signals operative to monitor systems and control various vehicle actuators. The vehicle control module **206** may include a steering controller for vehicle lateral motion control and a propulsion and braking controller for vehicle longitudinal motion.

(29) The road signal module **230** may dispense paint onto a surface. For example, the perception module **202** may detect an event in the surrounding environment or an abnormality on a roadway (e.g., a crash, temporary constructs, a construction site, etc.). The perception module **202** may communicate data associated with the event to the road signal module **230**. The road signal module **230** may determine an event type of the event based on the data. For example, the data may be image or video data of the event. The road signal module **230** may analyze the data using one or more techniques (e.g., neural networks, image processing techniques) to determine the event type. Event types may include temporal designations (e.g., a short duration event, a long duration event, etc.), distance designations (e.g., feet, yards, miles, away), characteristic designations (e.g., construction zone, road debris, etc.), emergency designations (e.g., high emergency, low emergency), or any other categorization of event type.

(30) Responsive to determining the type of event, the road signal module **230** may select a paint dispenser from a set of paint dispensers. For example, a first subset of paint dispensers may be associated with indicating a duration of the event. A first color of the first subset may indicate the event is predicted to last for a short period of time (e.g., 1 hour). A second color of the first subset may indicate the event is predicted to last for a long period of time (e.g., multiple days). In some examples, the first color and the second color may disappear (e.g., dissolve, become undetectable by a sensor) after the respective period of time has elapsed. A second subset of paint dispensers may be associated with communicating the type of event. For example, each type of event may be

associated with a respective type of paint. In some cases, how the paint dispenser dispenses the paint determines the information communicated to other vehicles. For example, the paint dispenser might communicate the duration of the event by the color of the paint and the event type by the shape, mark, or pattern of the paint made while dispensing the paint. In some cases, the paint may be dispensed to create a barcode (e.g., a quick response (QR) code) on the surface of the road, or dispense another symbol to convey information.

(31) In some cases, the road signal module **230** may communicate with a database. For example, the database may be a local database stored in memory **214**, a remote database stored on a remote computer or stored on a cloud network, or partially stored on a combination thereof. The database may include a mapping between the event types and distance to the event and different types of paint and/or designs. For example, each paint may include respective characteristics (e.g., color, wavelength, durability). Based on the respective characteristics, paint may be selected to represent different types of events and distances to the event. For example, a car crash may be associated with a first type of paint, a construction zone may be associated with a second type of paint, a distance of 100 meters to the car crash may be associated with the first type of paint, and a distance of 200 meters to the car crash may be associated with a third type of paint. In some cases, a design (e.g., shape, mark, pattern) may be selected in addition to the paint type to communicate additional information. In some cases, the paint type and/or the designs may indicate a type of event, a duration of the event, a distance to the event, navigational information (e.g., a location, a speed limit, other information that can be included on a road sign), or a navigational command (e.g., merge into another lane, reduce speed), among other information. For example, the car crash may be associated with a first design and the distance to the car crash may be associated with different types of paint (e.g., a first type of paint dispensed according to the first design may indicate 100 meters to a car crash and a second type of paint dispensed according to the first design may indicate 200 meters to the car crash).

(32) In some examples, a second vehicle **102** may detect the paint on the road. The second vehicle **102** may detect the paint (e.g., via a respective perception module **202**) before detecting the event. For example, the second vehicle **102** may include a sensor (e.g., a multi-spectral sensor, a video or image capturing device, etc.). The second vehicle **102** may monitor the road ahead of the second vehicle **102** and detect the paint. The second vehicle **102** may query a database and determine information associated with the paint. For example, the information may include a type of the event, navigational information, or a navigational command. The second vehicle **102** may determine a navigational action to perform based on the information. In some cases, the second vehicle **102** may update the paint. For example, the second vehicle **102** may dispense same or different second paint on the road based on changes to the event. The second vehicle **102** may dispense the second paint at a distance further from the event than the first paint to support communication of the event to other vehicles **102** earlier.

(33) FIG. 3 is a bird's-eye view of a roadway that supports dispensing road markings outside of a visible spectrum, according to an embodiment. FIG. 3 illustrates an environment **300** that includes a vehicle **302** and a roadway **306**. The vehicle **302** can include one or more sensors **308**, an autonomy system **304**, and paint dispensers **314** (e.g., located below the vehicle **302**, on a surface of the vehicle **302**). The vehicle **302** can be the same as or similar to the vehicle **102**. The roadway **306** can include a surface **310** and an event (e.g., abnormality) **312**.

(34) In some cases, the vehicle **302** may monitor (e.g., collect data regarding) the environment **300**. The vehicle **302** may detect, via the sensors **308** (e.g., a range of multi-spectral sensors), the event **312**. In some cases, the event **312** can be a construction zone, a police barrier, or other type of event. The vehicle **302** may determine the event type based on the collected data. For example, the vehicle **302** may utilize one or more techniques (e.g., image processing, neural networks, etc.) to determine the type of the event **312**. In some cases, the vehicle **302** may query a database responsive to the determination of the event type. The database may include event types

corresponding to types of paint, designs (e.g., patterns, shapes, amount of paint), barcodes (e.g., QR codes), etc.

(35) Based on the event type, the vehicle **302** may select one or more paint dispensers **314** to dispense paint. For example, the vehicle **302** may select a first type of paint including a first set of characteristics corresponding to the event **312**. In one example, the event **312** is a crash. The vehicle **302** may determine an approximate period of time the crash is expected to last (e.g., based on the database or based on one or more calculations). The vehicle **302** may select a paint **316** reflective in the infrared spectrum to indicate the period of time, a design (e.g., a line, a symbol, a mark, a pattern), and a distance between to the crash to indicate the event type. Advantageously, by reflecting in a non-visible wavelength, the paint **316** may avoid interfering with other existing road markings (e.g., for human operated vehicles). In some implementations, the first set of characteristics may include reflectivity, wavelength (e.g., spectral bands), color, durability, design, or other types of characteristics. Each characteristic may correspond to a distance, duration of time, navigational information, navigational command, or other type of information. Responsive to selecting the one or more paint dispensers **314**, the vehicle **302** may dispense the paint **316** onto the surface **310** of the road **306**. In some examples, the vehicle **302** may dispense the paint **316** below, in front of, or behind the vehicle **302** while driving on the road **306**. The vehicle **302** may perform a navigation action (e.g., merge into another lane) in response to dispensing the paint **316** and detecting the event **312**.

(36) In some cases, based on the event type, the paint **316** may disappear after a period of time has elapsed. Various embodiments of the paint **316** may remain on the surface **310** of the road **306** for varying durations. The vehicle **302** may select the type of paint based on an expected duration of the type of the event **312** encountered. For example, the event **312** may be a construction or repair zone that is expected to last for an entire day. The event **312** may be an accident (e.g., a crash) that is expected to last for multiple hours. The event **312** may be a police pull-over that is expected to last for multiple minutes. For each example, the paint **316** may disappear after the respective period of time has elapsed. In some cases, an emergency vehicle (e.g., an ambulance, a firetruck, a police vehicle, or a construction vehicle) may dispense the paint in response to establishing the event **312** or arriving to the event **312**. In some cases, the paint dispensers **314** may be a part of a hand-held system that may be operated by forward observers of a military ground force to mark routes for incoming troops. For example, by selecting a reflective waveband and adjusting a duration of the paint the paint may result in increased security in signaling to the incoming troops (e.g., hard to detect by enemy troops, unknown code).

(37) FIG. 4 is a bird's-eye view of a roadway that supports dispensing road markings outside of a visible spectrum, according to an embodiment. FIG. 4 illustrates an environment **400** that includes a vehicle **402** and a roadway **406**. The vehicle **402** can include one or more sensors **408**, an autonomy system **404**, and paint dispensers **414** (e.g., located below the vehicle **402**, on a surface of the vehicle **402**). The vehicle **402** can be the same as or similar to the vehicle **102**. The roadway **406** can include a surface **410** and paint (e.g., a marking) **412**.

(38) In some cases, the vehicle **402** may monitor (e.g., collect data regarding) the road **406**. The vehicle **402** may detect, via the sensors **408** (e.g., a range of multi-spectral sensors), the paint **412**. For example, the paint **412** may correspond to an event (e.g., the event **312**). The vehicle **402** may detect the paint **412** before detecting (e.g., before being in proximity to) the event. The paint **412** may include one or more characteristics. For example, the paint **412** may be reflective in the non-visible spectrum (e.g., may be reflective in a first spectral band of the non-visible spectrum). The paint **412** may be of a type of durability (e.g., soluble) that will disappear after a period of time after being dispensed has elapsed. The paint **412** may be configured according to a type of design (e.g., a QR code, a line, a shape, a symbol).

(39) The vehicle **402** may determine information associated with the event based on the one or more characteristics of the paint **412**. The vehicle **402** may query a database (e.g., a local database,

a remote database) based on the one or more characteristics. The query may include an indication of the one or more characteristics. The database may include information (e.g., a distance to the event, a type of the event, a duration of the event, navigational information, or one or more navigational commands) associated with the event corresponding to the one or more characteristics. Based on the information, the vehicle **402** may determine a navigational action to perform. For example, the event may be a crash in a first lane that the vehicle **402** is currently driving. Advantageously, the vehicle **402** may determine to merge into a second adjacent lane to avoid congestion caused by the crash in the first lane.

(40) In some cases, responsive to detecting the paint **412**, the vehicle **402** may dispense second paint **416**. For example, the paint **412** may be located a first distance away from the event. To increase communication to other vehicles, the vehicle **402** may dispense the second paint **416** at a location at a second distance further than the first distance away from the event. Other vehicles may detect the second paint **416** and perform a navigational action earlier (in relation to arriving at the event) than if they detected the first paint **412**. In some cases, the vehicle **402** may dispense the second paint **416** different from the first paint **412**. For example, based on a duration since the first paint **412** was dispensed, the vehicle **402** may calculate (e.g., based on a difference between the duration and the current elapsed time) a second duration before the first paint **412** disappears. The second duration can be the amount of time until the first paint disappears. The vehicle **402** can determine the current elapsed time using machine learning techniques, for example, or any other techniques based on characteristics of the paint **412** (e.g., an amount of fading in the paint **412**, which can be depicted in images or other sensor data that capture the paint **412**). The vehicle **402** may dispense the second paint **416** with a characteristic to disappear in accordance with the second duration. In some cases, the vehicle **402** may select the second paint **416** with a characteristic corresponding to the second distance away from the event.

(41) FIG. 5 shows execution steps of a processor-based method using the system **250**, according to some embodiments. The method **500** shown in FIG. 5 comprises execution steps **502-512**. However, it should be appreciated that other embodiments may comprise additional or alternative execution steps, or may omit one or more steps altogether. It should also be appreciated that other embodiments may perform certain execution steps in a different order. Steps discussed herein may also be performed simultaneously or near-simultaneously.

(42) FIG. 5 is described as being performed by a data processing system stored or on or otherwise located at an autonomous vehicle, such as the autonomous vehicle **102**, **302**, or **402** depicted in FIGS. 1, 3, and 4. However, in some embodiments, one or more of the steps may be performed by a different processor, server, or any other computing feature. For instance, one or more of the steps may be performed via a cloud-based service or another processor in communication with the processor of an autonomous vehicle and/or the autonomy system of such an autonomous vehicle.

(43) At **502**, the data processing system collects, using a sensor of the autonomous vehicle, data regarding an environment surrounding the autonomous vehicle as the autonomous vehicle is moving on a surface. At **504**, the data processing system determines whether an event has occurred in the surrounding environment from the data collected. For example, the data processing system can detect the event based on processing (e.g., according to one or more processing techniques) image or video data obtained from the sensor (e.g., a video or image device, camera, LiDAR device). In some embodiments, the data processing system can detect an abnormality on a road in the surrounding environment. For example, the abnormality may be debris located on the road in the surrounding environment. In some cases, at **505**, the data processing system determines whether paint (e.g., non-visible paint) is on a surface of a road from the data collected. For example, the data processing system can detect the paint on the surface of the road based on processing data obtained from the sensor (e.g., multi-spectral sensors).

(44) At **506**, if the data processing system detects the event, the data processing system determines an event type of the event based on the collected data. For example, the data processing system can

utilize image processing algorithms, machine learning models (e.g., one or more neural networks, support vector machines, random forests, etc.) or other techniques to process the data (e.g., image or video data). By utilizing such techniques, the data processing system can categorize the event into one of a set of event types. For example, the set of event types may include temporal designations (e.g., a short duration event, a long duration event, etc.), characteristic designations (e.g., construction zone, road debris, etc.), emergency designations (e.g., high emergency, low emergency), distance to the event designations, or any other categorization of event type.

(45) At **508**, the data processing system selects a paint dispenser from multiple paint dispensers based on the determined event type. For example, each dispenser can correspond to a different type of event and include paint of a different spectral band outside of a visible spectrum (e.g., within a non-visible spectrum). The data processing system can select a first paint dispenser that corresponds to the determined event type. In some cases, the paint dispensers can include sets of paint dispensers for each spectral band. Each paint dispenser of a set can correspond to a distance between a current location of the autonomous vehicle and events detected by the autonomous vehicle. For example, if the event is a first distance away from the autonomous vehicle (e.g., 500 feet), the data processing system may select a paint dispenser from the set that corresponds to the first distance. In some implementations, each paint dispenser of the multiple paint dispensers can correspond to a defined period of time in which the paint of the respective paint dispenser will remain on the surface (e.g., a period of time from dispensing the paint till the paint disappears). For example, if the event type corresponds to a short duration of time (e.g., one hour), the data processing system may select a paint dispenser that will remain on the surface for the short duration (e.g., that will disappear after the short duration expires). In some examples, the data processing system can select a paint to load into a paint dispenser (e.g., rather than selecting a paint dispenser, in addition to selecting a paint dispenser).

(46) At **510**, the data processing system automatically dispenses paint from the selected paint dispenser onto the surface. In some cases, the data processing system can dispense the paint according to one or more designs (e.g., patterns, shapes, QR codes). For example, the data processing system can query a database, the query including the event type. The database can include a mapping between event types and characteristics of types of paint, a design, and other information (e.g., navigational commands). The data processing system can obtain the characteristics and the design from the database and select the paint (e.g., the paint dispenser) that corresponds to the characteristics and dispense the paint onto the surface according to the design. At **512**, the data processing system performs a navigational action based on the detected event.

(47) In some implementations, the data processing system may detect a second event in the environment. The data processing system can determine a second event type of the second event and determine a second period of time associated with the second event type that is different from a first period of time associated with the first event type. The data processing system can select a second paint dispenser, different from the first paint dispenser, based on the second period of time and automatically dispense second paint associated with the second period of time from the selected second paint dispenser onto the surface.

(48) At **506**, if the data processing system detects paint on the surface of the road (e.g., first paint, second paint detected after dispensing first paint on the surface of the road), the data processing system determines the event type of the paint based on determining one or more characteristics of the paint. For example, the one or more characteristics can include a first spectral band of the detected paint. The first spectral band may be of a non-visible spectrum. For example, a standard may separate the non-visible spectrum into multiple bands (e.g., ranges of wavelengths). The paint may reflect a wavelength that is within a first spectral band of the standard. The first spectral band may correspond to an event type, navigational commands, a distance to the event, a duration of time corresponding to the event, or other information.

(49) At **508**, responsive to detecting the paint (e.g., second paint), the data processing system

selects a paint dispenser (e.g., a second paint dispenser) based on the first spectral band of the paint. In some cases, the paint dispenser corresponds to a second spectral band different from the first spectral band. For example, the data processing system can determine the first spectral band to correspond to a first period of time, indicating that the paint will disappear after the first period of time elapses. The data processing system can determine a second period of time associated with a difference between the first period of time and an elapsed time period (e.g., an amount of time that the paint has been on the road). The second spectral band may correspond to the second period of time (e.g., will disappear after the second period of time elapses). In this way, the detected paint and the selected paint may disappear at the same time (e.g., approximately the same time). At **510**, the data processing system automatically dispenses paint from the selected paint dispenser (e.g., the second paint dispenser) onto the surface.

(50) At **512**, the data processing system performs a navigational action based on the detected paint. For example, the data processing system can query a database, the query including an indication of the detected paint. The database can include a mapping between paints (e.g., characteristics of paints) and navigational commands, event types, distance to the event, a period of time the event is approximated to last, or other information. The data processing system can obtain the information from the database and determine a navigational action based on the information. For example, the data processing system may obtain a command to merge into another lane based on the first spectral band of the paint. The data processing system may control the autonomous vehicle to merge the autonomous vehicle into another lane based on the command. In some implementations, the data processing system can select the navigational action from multiple navigational actions based on the first spectral band.

(51) The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various components, blocks, modules, circuits, and steps have been generally described in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of this disclosure or the claims.

(52) Embodiments implemented in computer software may be implemented in software, firmware, middleware, microcode, hardware description languages, or any combination thereof. A code segment or machine-executable instructions may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc., may be passed, forwarded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, etc.

(53) The actual software code or specialized control hardware used to implement these systems and methods is not limiting of the claimed features or this disclosure. Thus, the operation and behavior of the systems and methods were described without reference to the specific software code being understood that software and control hardware can be designed to implement the systems and methods based on the description herein.

(54) When implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable or processor-readable storage medium. The steps of a method or algorithm disclosed herein may be embodied in a processor-executable software module, which may reside on a computer-readable or processor-readable storage medium. A non-transitory computer-readable or processor-readable media includes both computer storage media and tangible

storage media that facilitate transfer of a computer program from one place to another. A non-transitory processor-readable storage media may be any available media that may be accessed by a computer. By way of example, and not limitation, such non-transitory processor-readable media may comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other tangible storage medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer or processor. Disk and disc, as used herein, include compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc, where “disks” usually reproduce data magnetically, while “discs” reproduce data optically with lasers.

Combinations of the above should also be included within the scope of computer-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product.

(55) The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the embodiments described herein and variations thereof. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the principles defined herein may be applied to other embodiments without departing from the spirit or scope of the subject matter disclosed herein. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

(56) While various aspects and embodiments have been disclosed, other aspects and embodiments are contemplated. The various aspects and embodiments disclosed are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. An autonomous vehicle, comprising: a sensor; a plurality of paint dispensers, each dispenser corresponding to a different type of event and comprising paint of a different spectral band, the different spectral band corresponding to the different type of event, wherein paint from each of the plurality of paint dispensers is outside of a visible spectrum; and one or more processors configured to: collect, using the sensor, data regarding an environment surrounding the autonomous vehicle as the autonomous vehicle is moving on a surface; detect an event in the environment surrounding the autonomous vehicle from the collected data; determine an event type of the event based on the collected data; select a paint dispenser from the plurality of paint dispensers based on the determined event type; and automatically dispense paint from the selected paint dispenser onto the surface.
2. The autonomous vehicle of claim 1, wherein the one or more processors are configured to: detect, via the sensor, second paint on the surface; and perform a navigational action based on the detection of the second paint.
3. The autonomous vehicle of claim 2, wherein: the second paint comprises a first spectral band, and the autonomous vehicle selects the navigational action from a plurality of navigational actions based on the first spectral band.
4. The autonomous vehicle of claim 1, wherein the one or more processors are configured to detect the event by: detecting an abnormality on a road in the environment surrounding the autonomous vehicle.
5. The autonomous vehicle of claim 1, wherein the one or more processors are configured to: detect second paint on the surface; responsive to detecting the second paint, select a second paint dispenser based on a spectral band of the second paint; and automatically dispense third paint from the selected second paint dispenser onto the surface.

6. The autonomous vehicle of claim 1, wherein the plurality of paint of dispensers comprises a set of paint dispensers for a spectral band, each paint dispenser of the set corresponding to a distance between a current location of the autonomous vehicle and events detected by the autonomous vehicle.

7. The autonomous vehicle of claim 1, wherein the one or more processors are configured to: detect second paint on the surface; determine one or more characteristics of the detected second paint, the one or more characteristics comprising a first spectral band of the detected second paint; select a second paint dispenser from the plurality of paint dispensers that corresponds to a second spectral band based on the first spectral band of the detected paint; and automatically dispense third paint from the selected second paint dispenser onto the surface.

8. The autonomous vehicle of claim 1, wherein the paint of the plurality of paint dispensers is configured to indicate the event type, navigational commands, distance to the event, or any combination thereof.

9. The autonomous vehicle of claim 1, wherein each paint dispenser corresponds to a defined period of time in which paint of the respective paint dispenser will remain on the surface.

10. The autonomous vehicle of claim 1, wherein the event type is associated with a first period of time, the one or more processors are further configured to: collect, using the sensor, second data regarding the environment surrounding the autonomous vehicle; detect a second event in the environment surrounding the autonomous vehicle from the collected second data, wherein the second event is of a second event type; determine a second period of time associated with the second event type, wherein the second period of time is different from the first period of time associated with the event; select a second paint dispenser, different from the paint dispenser, from the plurality of paint dispensers based on the second period of time; and automatically dispense second paint associated with the second period of time from the selected second paint dispenser onto the surface.

11. A method for dispensing paint onto a surface, comprising: collecting, by at least one processor using a sensor, data regarding an environment surrounding a vehicle as the vehicle is moving on the surface; detecting, by the at least one processor, an event in the environment surrounding the vehicle from the collected data; determining, by the at least one processor, an event type of the event based on the collected data; selecting, by the at least one processor, a paint dispenser from a plurality of paint dispensers based on the determined event type, each dispenser of the plurality of paint dispensers corresponding to a different type of event and comprising paint of a different spectral band, the different spectral band corresponding to the different type of event, wherein paint from each of the plurality of paint dispensers is outside of a visible spectrum; and automatically dispensing, by the at least one processor, paint from the selected paint dispenser onto the surface.

12. The method of claim 11, further comprising: detecting, via the sensor, second paint on the surface; and performing a navigational action based on the detection of the second paint.

13. The method of claim 12, wherein: the second paint comprises a first spectral band, and the vehicle selects the navigational action from a plurality of navigational actions based on the first spectral band.

14. The method of claim 11, wherein detecting the event further comprises: detecting an abnormality on a road in the environment surrounding the vehicle.

15. The method of claim 11, further comprising: detecting second paint on the surface; responsive to detecting the second paint, selecting a second paint dispenser based on a spectral band of the second paint; and automatically dispensing third paint from the selected second paint dispenser onto the surface.

16. A controller comprising: one or more processors, the one or more processors configured to: collect, using the sensor, data regarding an environment surrounding an autonomous vehicle as the autonomous vehicle is moving on a surface; detect an event in the environment surrounding the autonomous vehicle from the collected data; determine an event type of the event based on the

collected data; select a paint dispenser from a plurality of paint dispensers based on the determined event type, each dispenser of the plurality of paint dispensers corresponding to a different type of event and comprising paint of a different spectral band, the different spectral band corresponding to the different type of event, wherein paint from each of the plurality of paint dispensers is outside of a visible spectrum; and automatically dispense paint from the selected paint dispenser onto the surface.

17. The controller of claim 16, wherein the one or more processors are configured to: detect, via the sensor, second paint on the surface; and perform a navigational action based on the detection of the second paint.

18. The controller of claim 17, wherein: the second paint comprises a first spectral band, and the autonomous vehicle selects the navigational action from a plurality of navigational actions based on the first spectral band.

19. The controller of claim 16, wherein the one or more processors are configured to detect the event by: detecting an abnormality on a road in the environment surrounding the autonomous vehicle.

20. The controller of claim 16, wherein the one or more processors are configured to: detect second paint on the surface; responsive to detecting the second paint, select a second paint dispenser based on a spectral band of the second paint; and automatically dispense third paint from the selected second paint dispenser onto the surface.
